

Kavli Institute for Particle Astrophysics and Cosmology
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Research Interests

3D multiphase ISM structure; Galactic magnetism; CMB foreground; Astrostatistics

Education

Ph.D., Physics, Stanford University Sept 2021 – Present

Advisor: Susan Clark

Thesis: Morphological Mapping of the Multiphase, Magnetized ISM

B.Sc., Physics & Math, University of Michigan Sept 2016 – June 2020

Advisor: James Wells

Thesis: “*Probing non-standard neutrino interactions with supernova neutrinos at Hyper-K.*”

Publications

In Prep

1. **Minjie Lei** et al., “*Polarized Dust Foreground Super-Resolution with Vision Transformer-based Technique*,” 2026, In prep
2. **Minjie Lei** et al., “*Imprints of Neutral Medium Complexity on Polarized Synchrotron Emission and Faraday Rotation: Implications for Galactic Magnetic Field Structure*,” 2026, In prep

Selected First-Author

1. ***Minjie Lei**, S. E. Clark, et al., “*Neutral gas phase distribution from HI morphology: phase separation with scattering spectra and variational autoencoders*,” *The Astrophysical Journal*, vol. 993, no. 4, 2025
2. **Minjie Lei**, S. E. Clark, “*A New Constraint on the Relative Disorder of Magnetic Fields between Neutral ISM Phases*,” *The Astrophysical Journal*, vol. 972, no. 1, 2024
3. **Minjie Lei**, S. E. Clark, “*Probing the cold neutral medium through HI emission morphology with the scattering transform*,” *The Astrophysical Journal*, vol. 947, no. 2, 2023

Other Publications

1. Rebecca Booth, Anna Ordog et al. including **Minjie Lei**, “*A Three-Dimensional Model for the Reversal in the Local Large-Scale Interstellar Magnetic Field*,” 2025, [arXiv:2512.03332](https://arxiv.org/abs/2512.03332), Accepted by ApJ
2. **Minjie Lei**, K. V. Tsang, et al., “*Implicit Neural Representation as a Differentiable Surrogate for Photon Propagation in a Monolithic Neutrino Detector*,” 2022, [arXiv:2211.01505](https://arxiv.org/abs/2211.01505)
3. **Minjie Lei**, Noah Steinberg, & James D. Wells, “*Probing non-standard neutrino interactions with supernova neutrinos at Hyper-K*,” 2020, *JHEP*, 01, 179
4. **Minjie Lei**, James D. Wells, “*Minimally modified A_4 Altarelli-Feruglio model for neutrino masses and mixings and its experimental consequences*,” 2020, *Phys. Rev. D*, 102 (1), 016023

Honors and Awards

2025–2026	2MB Graduate Fellowship: Awarded to a senior graduate student who has done innovative work using statistical techniques applied to astrophysics, Stanford University
2023–2025	Stanford Graduate Fellowship: Awarded to graduate students in science and engineering who has demonstrated excellence in doctoral-level research and study, Stanford University
2017–2020	James B. Angell Scholar, University of Michigan

Scientific Presentations

Invited Talks	
Oct 2025	Galaxy Seminar ; Flatiron CCA
Sep 2025	Low Density Universe Meeting ; STScI
Sep 2025	SFIR Seminar ; Princeton Astro
May 2025	Astronomy Tea Talk ; Caltech
April 2025	ITC Luncheon ; Harvard CfA
March 2025	Scattering Club ; ENS Paris
Nov 2024	Astronomique Series ; Ciela Institute
March 2024	KIPAC tea ; Stanford University
Jan 2024	Journal Club ; Pan-Experiment Galactic Science Group
February 2023	ISM Seminar ; ENS Paris
Contributed Talks	
Jan, 2026	AAS 247 ; Phoenix, Arizona
July, 2025	Interstellar Institute 7 ; Institut Pascal - Université Paris-Saclay
Feb, 2025	Structure and polarization in the interstellar medium ; Stanford University
Jun. 2024	AAS 244 ; Madison, Wisconsin
July. 2023	Interstellar Institute 6 ; Institut Pascal - Université Paris-Saclay
Dec. 2022	Galactic Science and CMB Foregrounds ; Core to Core CMB Workshop series

Teaching and Mentorship

Teaching Assistantships	
Spring 2024	Computational Physics (Physics 113)
Fall 2023	Stars and Planets in a Habitable Universe (Physics 15)
Winter 2023	Stars and Planets in a Habitable Universe (Physics 15)
Fall 2019	Honors Physics I - Mechanics (Physics 160)
Fall 2018	Physics for the Life Sciences I (Physics 135)
Student Mentorship	
2024-present	Caio Gould , undergraduate at Stanford
2023-2024	Abraar Saleem , Stanford postbac scholar, Stanford KIPAC postbac program
2022-2023	Amritpal Nijjar , undergraduate at UCLA, Stanford PIE Program
2020-2021	Jackie Zhao , High school student, Minds Matter Philadelphia

Service and Outreach

2025	Journal Reviewer : A&A
2025	Booth Leader & Volunteer ; KIPAC Community Day, Stanford University
2025	Local Organizing Committee ; Conference in Honor of Prof. John Dickey, Stanford University
2024	Booth Leader & Volunteer ; STEMfest, Stanford University
2023	Volunteer ; KIPAC Community Day, Stanford University

References

Susan Clark, Stanford University
Erwan Allys, ENS Paris
Doug Finkbeiner, Harvard University

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